

Development and Testing of the Re-Deployable Radiator for Deep Space Exploration Technology Demonstrator, DESTINY⁺

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This paper reports the design, fabrication, and testing of a reversible thermal panel breadboard model (RTP-BBM). RTP is a flexible, re-deployable radiator that autonomously controls the temperature of a heat source. It promotes heat dissipation by deploying the radiator surface when the heat source is at a high temperature. Conversely, in a cold case, heat dissipation is conserved by stowing the radiator surface. This study conducts a deployment/stowing test and thermal vacuum test to evaluate the validity of the design, and model correlations are conducted via thermal analysis. The RTP-BBM comprises high-thermal-conductive graphite sheets as the flexible fin, and shape-memory alloys (SMA) as a temperature sensitive passive actuator. The deployment/stowing test was conducted in a thermal constant bath, where it was confirmed that the fin was deployed and stowed according to the SMA temperature. However, temperature hysteresis of up to +60°C was confirmed between heating and cooling cycles. In the thermal vacuum test, power step and power cycle tests were conducted. It was determined that the fin deployed and stowed according to the temperature of the onboard equipment while autonomously regulating the temperature. Additionally, the thermal analysis model correlated with the experimental results showing good agreement within ±6°C.

Nomenclature

A	=	area
Δt	=	time step
Q	=	heat amount
T	=	temperature
ε	=	emissivity

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σ = Stefan-Boltzmann constant

Subscripts

Rad = radiation

RS = radiator surface

S = shroud

I. Introduction

ALONG with the high performance of electronic equipment installed in spacecraft and the diversification of modern missions, the heat density of spacecraft has increased remarkably. Currently, in several spacecrafts, thermal control is regulated by both the heater and radiator to keep mounted electronic devices within the allowable temperature range. [1] If a change in the external thermal environment of the Earth orbiting satellite is small, this thermal control method can be excellent because of its reliability and simplicity. Conversely, it can also be considered to be a poorly efficient thermal control method owing to the increase in electric power used by the heater, as well as the increased radiator weight when the external thermal environment changes is large such as a planetary explorer. For spacecraft with limited electric power resources, electricity used for thermal control must be as small as possible; a low total load is preferable from the viewpoint of launch cost effectiveness.

With emphasis on qualities such as simplicity and functional design (low weight), attention has been focused on the Reversible Thermal Panel (RTP). The RTP is a flexible radiator that can promote heat dissipation by deploying a heat dissipation surface when a heat source is at high temperature, allowing for the suppression of heat dissipation via the stowing of the heat dissipation surface at a low temperature. The RTP is an autonomous thermal control device proposed by Nagano et al in 2002. [2] Figure 1 shows a schematic of RTP operation. A unique feature of RTP—through the use of a shape-memory alloy (SMA) for deployment and stowing—is the ability to handle a wide range of heat input without the need of a power source. Furthermore, for thermal diffusion in the radiator fin plane, weight reduction and simplification of the radiator are performed using a high-thermal-conductive material for the radiator fin itself. [3] [4] Conversely, it is currently the case that space demonstration cases have yet to be conducted.

Therefore, the first space demonstration of the RTP will occur through its installation on DESTINY+. DESTINY+ is a technology demonstrator used in the study of future low cost, high frequency, sustained deep space exploration. DESTINY+ plans to implement RTP as one of the advanced thermal control devices used to eliminate concentrated heat generation from the ion engine to ensure effective heat dissipation. [5] [6]

In this study, a deployment/stowing test and thermal vacuum test are performed on the RTP-BBM (currently being produced). We constructed a thermal analysis model using Thermal Desktop which is one of a thermal analysis software and performed an evaluation of its design validity and model correlation.

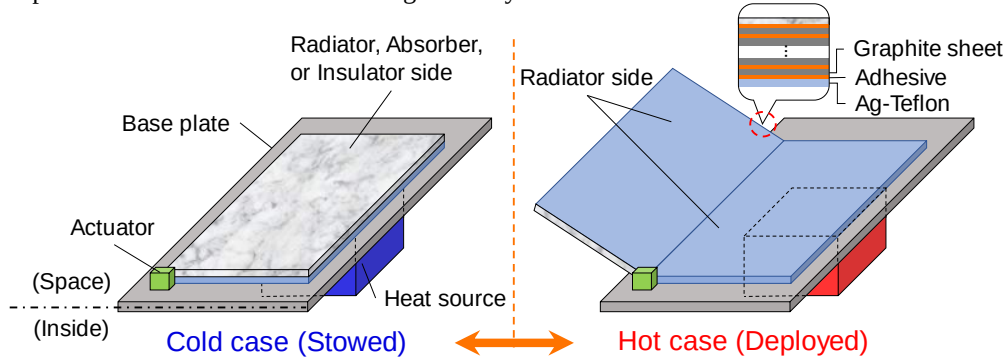


Figure 1 Schematic of RTP operation.

II. Experimental Device

In this study, a thermal performance test was performed on the RTP-BBM shown in Figure 2. The heat dissipation area was 240 mm × 600 mm at the time of stowing and 720 mm × 600 at the time of deployment. The RTP sometimes has heat absorption or heat retention characteristics at the time of radiator fin stowing due to the application of a material possessing a high solar absorptivity/emissivity ratio and the multilayer insulation (MLI) on the back of the fin.[2] [7] In contrast, the fin back side of this RTP is also designed to act as heat radiation surface so that a certain amount of heat release can be secured even upon stowing.

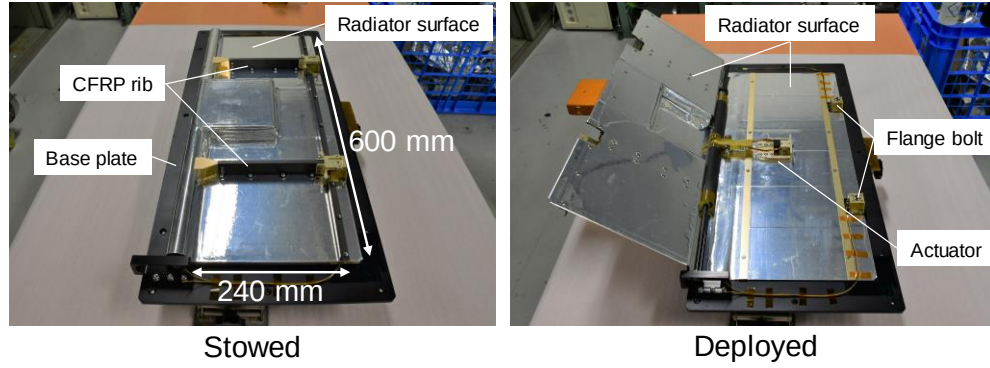


Figure 2 Appearance of RTP-BBM.

Figure 2 shows the appearance of the deployment/stowing mechanism. In the act of deployment and stowing the fin, the single crystal shape-memory alloy (SCSMA) based Cu–Al–Ni pushes out the drive crank and torsion bar to convert the horizontal force into a rotational force. Since the frame attached to the back surface of the fin is directly fixed to the torsion bar via bolts, the fin is opened and closed by rotating the torsion bar.

Since an actuator was installed on top of the radiator surface in the conventional RTP, there was an issue in that it was difficult to deploy and stow the fin completely due to the lack of torque of the actuator. In terms of temperature response, there is another problem in that the thermal resistance is large and the temperature response is poor due to the actuator being separated from the heat source. In order to solve the problem of the conventional RTP, our modified RTP uses an actuator with a newly designed and manufactured torsion bar. With the actuator positioned at the center of the radiator surface, this measure is expected to greatly improve the deployment/stowing performance of the fin and temperature response when compared to the conventional RTP.

The radiator panel is manufactured with 16 layers of Kaneka Graphite Sheets (XGX-040R) with thicknesses of 40 μm . The portion where the radiator panel is bent by the torsion bar is purposely not glued to maintain flexibility. However, the other layers are adhered with an adhesive. Silverized Teflon is applied to the front and rear surfaces, with the total thickness of the radiator panel measuring approximately 1.7 mm.

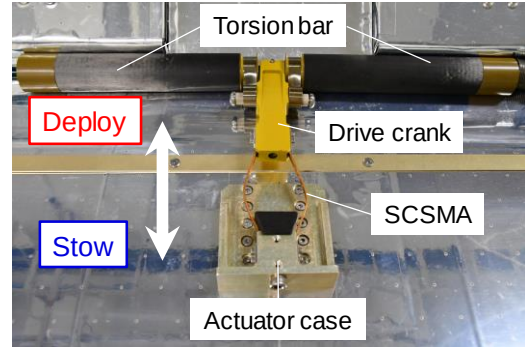


Figure 3 Appearance of the Actuator.

III. Deployment/Stowing Test

A. Experimental Setup

In order to grasp the deployment and stowing operation of the RTP under atmospheric environmental conditions, a deployment/stowing test was conducted in a thermal constant bath. The schematic layout of the experimental equipment is shown in Figure 4. Temperature measurements were conducted using a T-type thermocouple; with 2 points on the actuator case, 1 point on the base plate, and 1 point inside the thermal constant bath (total of 4 points). The deployment angle measurement was geometrically obtained using the images taken by a fixed point camera placed outside the thermal constant bath.

A temperature step test was conducted to confirm the successful basic operations. Temperature control was conducted by increasing the temperature from $-40\text{ }^{\circ}\text{C}$ to $+80\text{ }^{\circ}\text{C}$ at $10\text{ }^{\circ}\text{C}$ increments and then decreasing the temperature from $+80\text{ }^{\circ}\text{C}$ to $-40\text{ }^{\circ}\text{C}$ at $10\text{ }^{\circ}\text{C}$ increments. In order to confirm the reproducibility of the operational deployment, the same test was repeated four times.

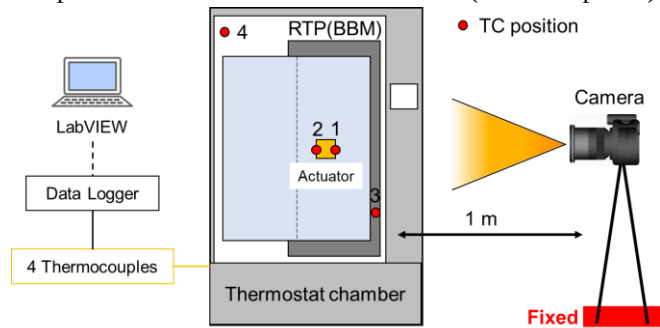


Figure 4 The schematic layout of the experimental equipment for deployment/stowing test.

A. Experimental Results

Figure 5 shows the relation between the development angle and actuator case temperature, while Figure 6 shows the deployment state of the RTP-BBM. In each experiment, it can be confirmed that the deployment angle of the radiator is 5° or less at the time of stowing and 150° at the time of deployment, and can be normally deployed and stowed. Regarding the reproducibility—although there are some differences when it comes to the temperature-raising operation, even when good agreement is observed during temperature-lowering operation—it is considered that reproducibility is generally confirmed.

It was confirmed that deployment initiated at $+20^\circ\text{C}$ during the temperature rise, gradually increased to $+50^\circ\text{C}$, increased again from $+50^\circ\text{C}$ to $+70^\circ\text{C}$ until the deployment reached the maximum deployment angle. Upon cooling, it was stowed dynamically at $+10^\circ\text{C}$ as it decreased to $+0^\circ\text{C}$, then the radiator fin gradually stowed at -10°C , until finally stowing at -20°C until completion. Although it is normal that the temperature hysteresis of operation occurs due to the nature of SMA, temperature hysteresis can be as large as $+50^\circ\text{C}$ to $+60^\circ\text{C}$. Since the allowable temperature range of the onboard equipment is approximately between -15°C and $+50^\circ\text{C}$, there is a possibility that deployment may not occur at high temperature. As a solution to this problem, it is necessary to reduce the temperature hysteresis by reexamining the length and width of the SMA, or by lowering the maximum deployment temperature.

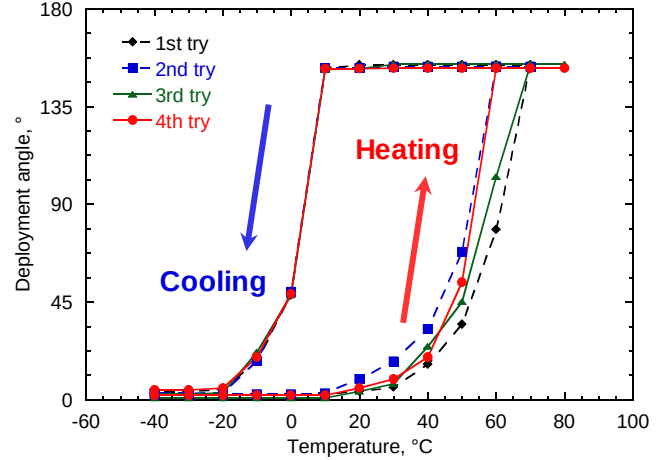


Figure 5 Relation between actuator temperature and deployment angle.

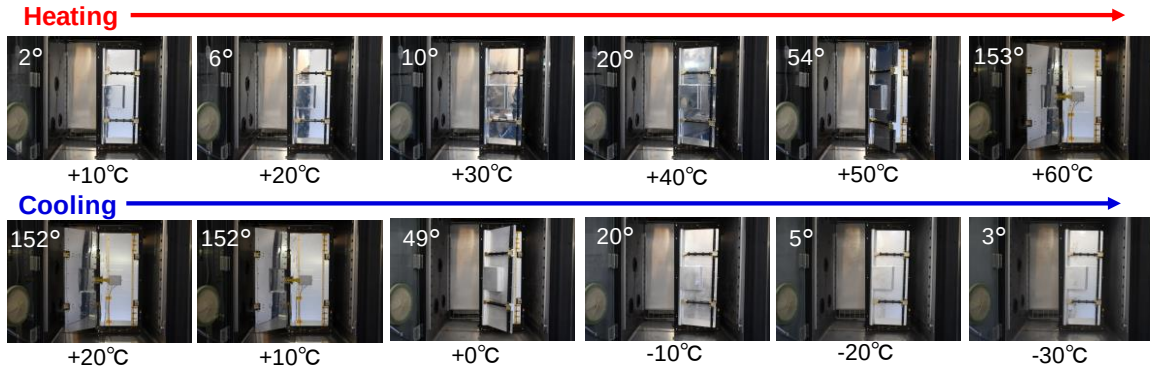


Figure 6 Deployment state during deployment/stowing test (Upper: Heating process, Lower: Cooling process).

IV. Thermal Vacuum Test and Thermal Analysis Model

A thermal vacuum test was conducted to verify the heat radiation performance of the RTP under a vacuum/cryogenic environment.

A. Experimental Setup

Figure 7 (a) shows the location of the heater during testing. The heat source was simulated by pasting seat heaters (200×600 mm) on the back of the aluminum base plate. The mounting position of the thermocouple is shown in Figure 7 (b) and (c). Temperature measurements were conducted using 42 T-type thermocouples, and heat insulation was applied using MLI, in which 10 layers were laminated on the base plate and the actuator for each.

The temperature of the actuator is required to be as close as possible to the heat source temperature to increase the temperature responsiveness of the fin's deployment and stowing operations. For this reason, an actuator cover was fabricated, with its front surface designed to be heat-insulating via the application of MLI and an aluminum deposition polyimide tape. Black Kapton tape was applied to the back surface to effectively equalize heat dispersion. The actuator cover is shown in Figure 7 (d).

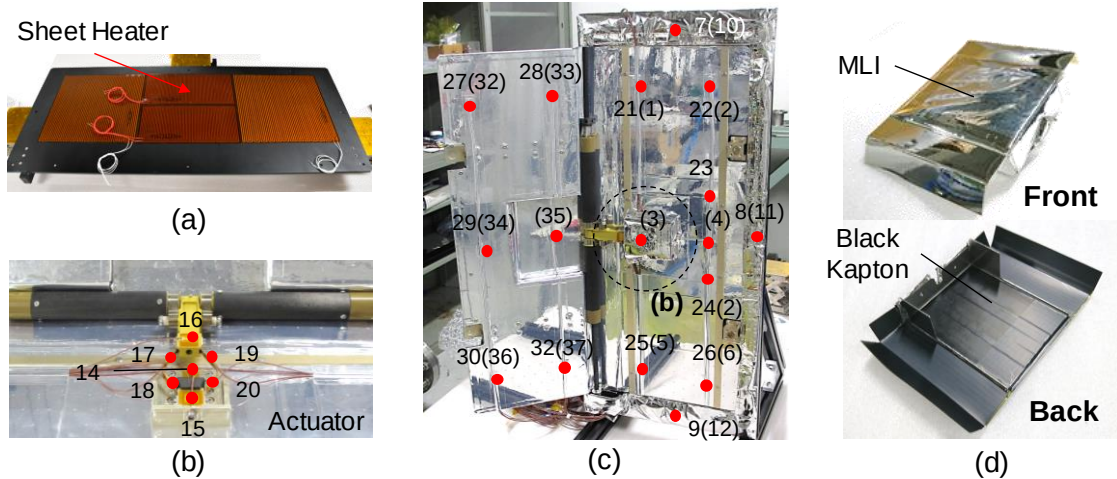


Figure 7 Experimental setup of the RTP for the thermal vacuum test; (a) is the heater location in the testing, (b) and (c) show the thermocouple positioning scheme on the actuator and entire RTP, (d) is the actuator cover with thermal insulation and the applied heat equalizer.

B. Thermal Vacuum Test System and Test Condition

The test was conducted in a thermal vacuum chamber to simulate a space-like environment. The shroud installed in the thermal vacuum chamber is cooled via liquid nitrogen, with the environmental testing temperature held at 100 K or less. Throughout the test, vacuum levels within the chamber were maintained at a pressure of 10^{-5} Torr. The temperature of each component is recorded at intervals of 3 seconds via the designated thermocouple attached to each part of the RTP. In addition, the development angle during the test was measured using images taken by a compact camera placed inside the thermal vacuum chamber, with photographs was taken at 10 second intervals. An example of the captured image is shown in Figure 8

In this experiment, three tests were conducted: a power step test, a power cycle test and a heat retention test. Table 1 lists the test conditions. In the power step test, we aimed to obtain the basic heat dissipation characteristics of the RTP by providing heat loads in the form of step-up and step-down adjustments. In the power cycle test, heat loads of 160 W and 40 W were alternately applied, and the response speed to large changes in the heat load were verified. In the heat retention test, it was verified how long it takes to reach the lower limit temperature (-40°C) when the heat input is lost from the high heat load state (160 W).

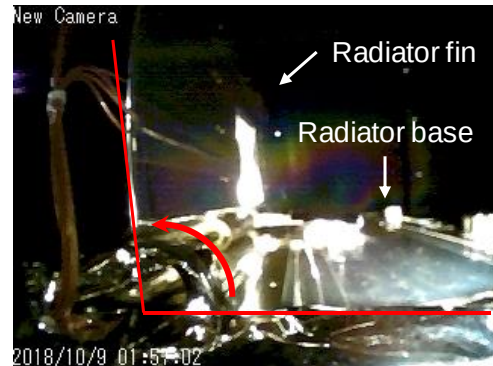


Figure 8 An example of the captured image of RTP in the thermal vacuum chamber.

Table 1 Test conditions.

Test	Applied power, W
Power step test	30-40-60-80-100-120-140-160-180-200-210-200-180-160-140-120-100-80-60-40-30
Power cycle test	100-160-40-160-40-160-100
Heat retention test	160-0-160

C. Thermal Analysis Model

A thermal analysis model of the RTP-BBM was constructed using Thermal Desktop (shown in Figure 9). The node size was up to 20×20 mm or less, and the radiator section was divided into approximately 10×10 mm sectors for higher resolution analysis. The environmental temperature was set at a shroud temperature of 77 K, with assumed correlation with the thermal vacuum test; the heat load was calculated in the range of 30 W to 210 W. A Kaneka

Graphite Sheet (high thermally conductive material) was used as the RTP radiator member. However, due to the sheet's large temperature dependence, the measured values of previously conducted research in this laboratory were used. The heat source size was 200×600 mm, which is the same heating area size used in the thermal vacuum test. In addition, the whole hemispheric emissivity of the radiator surface is made so by applying Silverized Teflon on the surface, including the back of the fin (0.77). The base plate is thermally insulated by MLI on both sides, and the total hemispheric emissivity of the surface and the effective emissivity of MLI were set to 0.63 and 0.05, respectively.

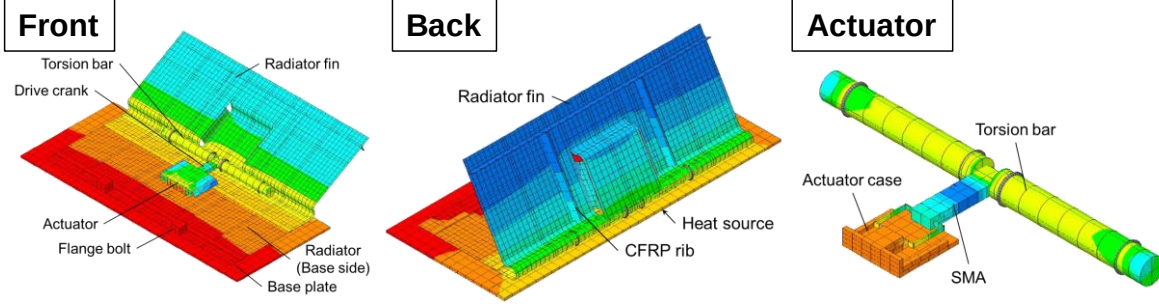


Figure 9 Thermal analysis model constructed using Thermal Desktop.

V. Test and Analytical Results and Discussions

A. Power Step Test Result

The temperature profile of the power step test is shown in Figure 10. This figure shows the temperatures in the power step test during power step-up and step-down increments of the thermal dummy, the radiator (base side), the radiator fin, the SMA, and the shroud. Focusing on the temperature of the radiator fin, it can be confirmed that the temperature change is gentle at 80 W in the step-up process, and at 100 W in the step-down process. This is a temperature change caused by the deployment and stowing of the fin, indicating the autonomous control of heat radiation.

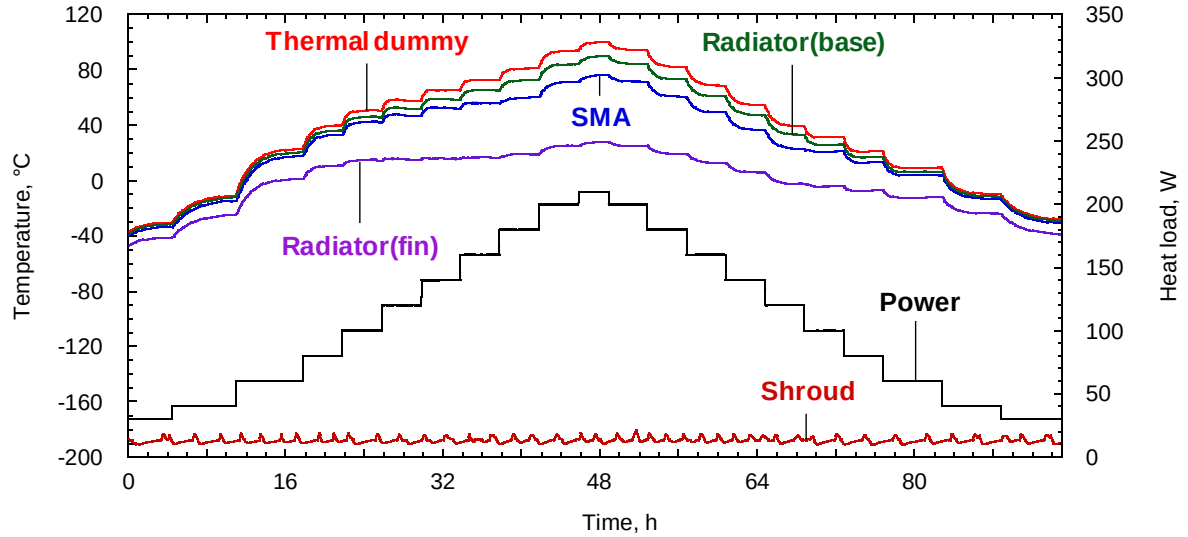


Figure 10 Temperature profile during power step test.

Next, the steady-state result of the thermal dummy temperature of the power step test is shown in Figure 11. During the step-up test, it is confirmed that the radiator fin begins to deploy at $+20$ °C as the heat radiation amount increases. During the step-down test, the radiator fin starts stowing at $+40$ °C as the heat radiation amount decreases. In addition, it was confirmed that the temperature at the end of deployment of this RTP was as high as $+80$ °C. This is due to the fact that the transition point temperature of SMA is too high, making it is necessary to lower the transition point temperature via SMA optimization.

In order to evaluate the validity of the thermal analysis model, we correlated it with the results of the thermal vacuum test. The contact heat transfer coefficient and the deployment angle for heat load were set as correlation targets, and the model was reconstructed and recalculated.

Figure 12 (a) shows the relation between the thermal dummy temperature and the heat rejection performance in the power step test. Focusing on the calculation results and experimental results at each deployment angle, it was confirmed that the thermal dummy temperature of the experimental result is lower than the 0° calculation result in the low temperature range. It is postulated that this is due to the fact that the fin could not be completely stowed in the low temperature range during the experiment (deployment angle being approximately 10°). In addition, when comparing the experimental result with the calculation result after the correlation, there is observed agreement with both results within $\pm 6^\circ\text{C}$, indicating that this analysis model is generally appropriate.

There are noticeable differences between both results in the high temperature range. It is postulated that the applied Silverized Teflon played a factor in this difference. In this test, a thermocouple is attached to the Silverized Teflon surface of the radiator, but due to the thermal resistance generated not less than a radiator surface and Silverized Teflon applied additionally, it is conceivable that the radiator surface temperature became higher in the experimental results than in the calculation result. Also, in the analytical model, factors such as failure to accurately model the fin shape as well as the incomplete correlation of the contact heat transfer coefficient may have also played a role in this result. Therefore, it may be necessary to further refine the analytical model and the contact heat transfer coefficient to provide a more accurate recalculation.

The experimental and calculation results of RTP temperature for each part are shown in Figure 12 (b). In the case of high heat load, temperature difference arises due to the factors mentioned above, but the temperature distribution of each part shows good agreement in the low/medium heat load region.

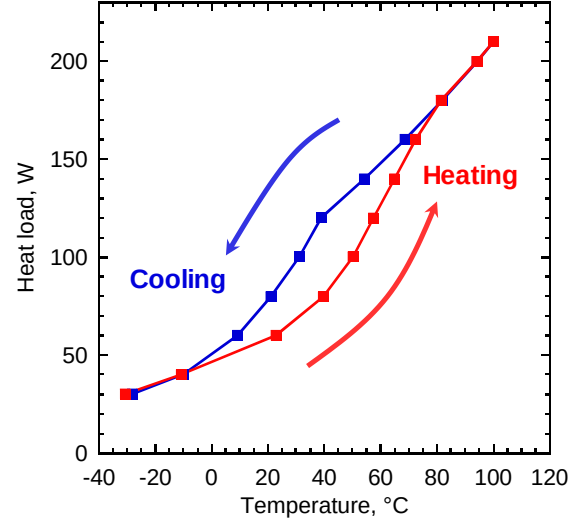


Figure 11 Relation between thermal dummy temperature and heat load at steady state during power step test.

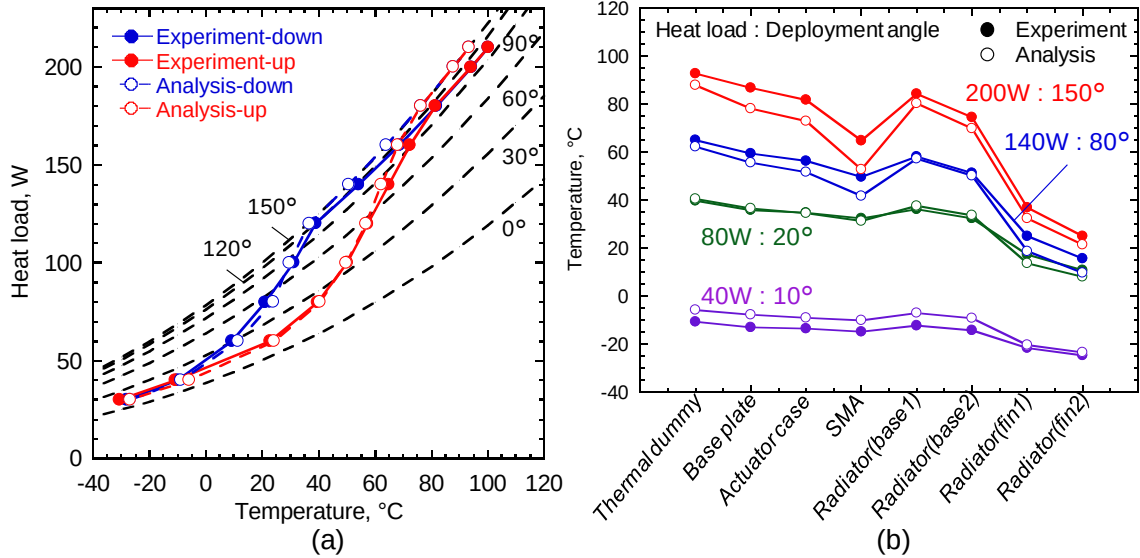


Figure 12 (a) Shows the relation between thermal dummy temperature and heat rejection capability during power step test. (b) Temperature of each RTP part by experiment and by calculation.

Furthermore, transient analysis was performed in a high heat load region (120 W to 210 W) where the deployment angle is constant at approximately 150°. The validity of the thermal analysis model was evaluated. The comparison of the thermal dummy temperature of the experimental result and analysis result is shown in Figure 13. The analysis results agree with the experimental results within 4 °C, and it is shown that the transient heat transfer behavior of the RTP-BBM can be evaluated using this model.

B. Power Cycle Test Result

The temperature profile of the power cycle test is shown in Figure 14. Notice that temperature rise is restrained at high temperature and temperature decline is suppressed at low temperature due to deployment and stowing of the radiator fin. We will verify this phenomenon by discussing response speed. The response speed was evaluated by verifying the time at which 65 °C varies from steady state (65 °C is 80% of the thermal dummy temperature difference in steady state between low temperature and high temperature). Here the thermal dummy temperatures at low temperature and high temperature are +72 °C and −9 °C, respectively, with a difference of 81 °C. As a result, it was confirmed that the time required for the cooling shift was 67 minutes and the time required for the heating shift was 41 minutes. The cooling shift took about 1.5 times longer than the heating shift. This is due to the fact that the heat release amount defined by the following equation (1) becomes larger at high temperature, so that the required time to reach thermal equilibrium is shortened.

$$Q_{rad} = \sigma \varepsilon A_{RS} \frac{T_{RS}^4 - T_S^4}{\Delta t} \quad (1)$$

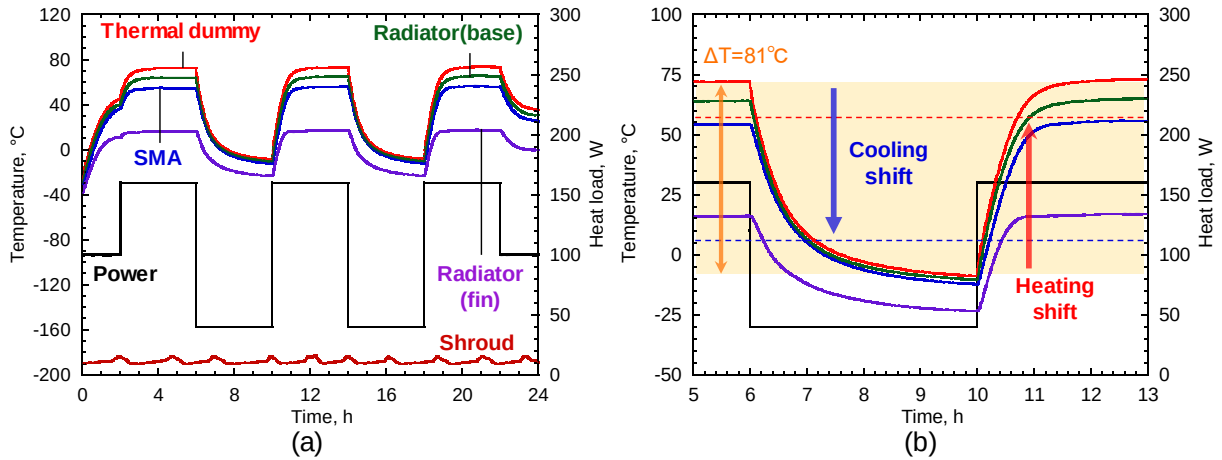


Figure 14 Temperature profile during power cycle test, (a) is a full timescale result and (b) is a local timescale result (5 hours to 13 hours).

C. Heat Retention Test Result

The experimental results of the heat retention test are shown in Figure 15 along with the results of the transient analysis at a deployment angle of 90°. The graph plots the thermal dummy temperature and the expansion angle of the experimental results, as well as the heat source temperature of the analysis results.

Comparing the high temperature shift of the analysis result with the experimental result, it can be confirmed that the response speed is faster in the experimental result. This is due to the deployment of the fin. The RTP's performance suggests that it is possible to reach thermal equilibrium more quickly than with general radiators. Also, focusing on the low temperature shift (the time to reach −40 °C, i.e., the lower limit temperature of RTP) was 1 hour and 18 minutes as depicted in the analysis results; whereas it took 1 hour and 50 minutes in the experimental results. Using these results, the RTP has shown that it is possible to extend the time in reaching the lower limit temperature by stowing the fin and keeping the mounted equipment warm.

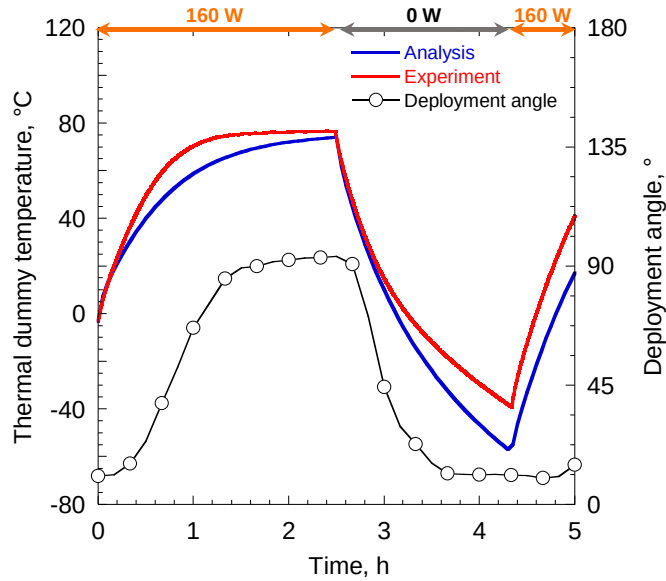


Figure 15 Temperature and deployment angle profile comparison of the analytical and experimental results during the heat retention test.

VI. Conclusion

A deployment/stowing test and thermal vacuum test were performed on the RTP-BBM, which is to be installed onboard the deep space exploration technology demonstrator, DESTINY+. The deployment operation and heat radiation performance were verified and correlated with a thermal analysis model. During the deployment/stowing test, temperature hysteresis was as high as 60 °C at maximum; in the thermal vacuum test, the temperature at the end of deployment was as high as 80 °C. Therefore, it was shown that temperature-transition optimization by way of SMA width/length shape adjustment (as well as the metal composition of SMA) is indispensable. The validity of the analysis model was also evaluated, showing that the experimental values and the calculated values nearly agree. In the future, it is important to improve the accuracy of this analysis model by refining the fin shape and setting the contact heat transfer coefficient with greater precision.

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